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Title: Design and Implementation of a Digital RPM Meter



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Design and Implementation of a Digital RPM Meter

Abstract—This project presents the design and implementation of a digital RPM (Revolutions Per Minute) meter using basic digital electronics components. The system employs an infrared (IR) sensor to detect the rotational speed of a motor shaft, which generates pulses corresponding to each revolution. A 555 timer-based monostable multivibrator is used to filter noise from the sensor signal, ensuring that only clean pulses are counted. Another 555 timer configured as a 6-second timer provides the time window for measurement. The outputs of the timer and the sensor are combined using an AND gate so that pulses are counted only within the measurement period. Three CD4026 counter ICs are used to drive a 7-segment display that shows the number of pulses detected. The RPM is obtained by multiplying the counter value by 10, as the 6-second measurement corresponds to one-tenth of a minute. The project demonstrates a simple, low-cost, and efficient technique for RPM measurement suitable for laboratory and small-scale motor applications..

Index Terms—RPM meter, IR sensor, 555 timer, CD4026, AND gate, digital counter

I. INTRODUCTION

ROTATIONAL speed measurement is essential in many engineering applications such as motor control, industrial automation, and testing of electromechanical systems. While commercial tachometers are available, they are often costly. This project aims to build a low-cost digital RPM meter using fundamental digital logic circuits for educational and practical applications.

Project Objectives

- To design an RPM meter using IR sensor, timer, and counters.
- To implement a noise-free sensor interface using a 555 monostable circuit.
- To enable time-based measurement using a 6-second timer.
- To display the RPM on a digital counter using CD4026 ICs.

Outline

This report is organized into several sections. The **Introduction** presents the background, motivation, and objectives of the project. The **Literature Review** summarizes related works that provide the foundation for this study. The **Methodology and Modeling** section describes the working principle of the RPM

meter, along with the details of components and experimental setup. The **Results and Discussion** section presents the experimental outcomes, cost analysis, and limitations of the system. Finally, the **Conclusion and Future Work** highlights the achievements of the project and suggests possible improvements. References are provided at the end, followed by appendices containing supporting diagrams.

II. LITERATURE REVIEW

- A 2020 study on **optical tachometers** highlighted the use of IR sensors for accurate speed detection in small motors.
- A 2021 paper on **555 timer applications** demonstrated the effectiveness of monostable multivibrators in noise filtering for digital counting circuits.
- In 2019, researchers proposed a **low-cost digital tachometer** using Arduino and sensors, showing the importance of simple pulse counting methods.
- A 2022 study on **digital counters** reviewed the use of CD4026 in 7-segment displays for measurement systems.
- In 2023, a paper presented an **AND gate-based gating technique** for enabling time-controlled pulse counting in RPM measurement.

III. METHODOLOGY AND MODELING

A. Introduction

The RPM meter design is based on pulse counting in a fixed time window.

B. Working Principle

The RPM meter operates by counting the number of revolutions within a fixed time interval and converting it to revolutions per minute. An infrared (IR) sensor is placed facing the rotating shaft, which generates a pulse for each revolution when a reflective strip passes in front of it. Since raw sensor signals may contain disturbances or multiple triggers, the output is first passed through a 555-timer configured in monostable mode, which filters noise and produces a single clean pulse per revolution. To ensure that pulses are counted only for a fixed duration, another 555 timer is used as a 6-second timer. The output of this timer and the filtered sensor pulses are fed into a 7408 AND gate. The AND gate ensures that the counter only receives pulses

IV Results and Discussion

1. **IR Sensor:** Detects shaft rotation.
2. **555 Timer (Monostable):** Used for pulse filtering and timer circuit.
3. **AND Gate:** Controls pulse flow to counter.
4. **CD4026 IC:** Drives 7-segment displays for count.
5. **7-Segment Display:** Displays pulse count (later scaled to RPM).

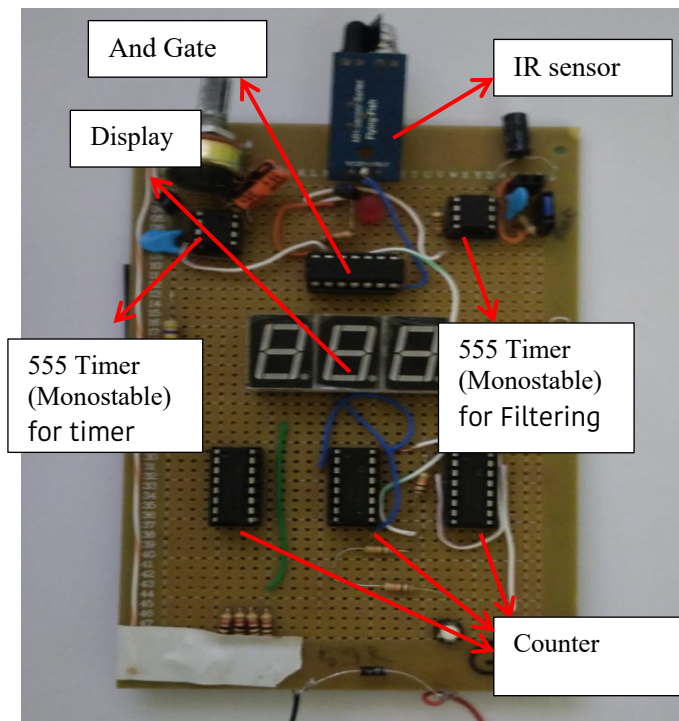


Figure 1 : Implementation of RPM meter

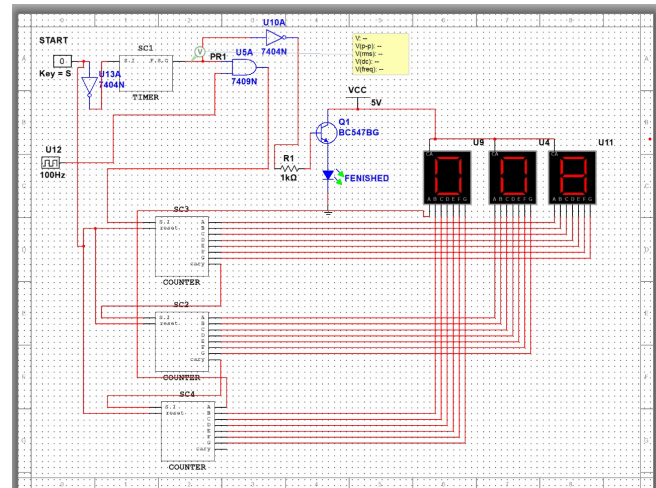


Figure 2: Simulation of the project

Although the circuit performed well, some limitations were observed. The accuracy of the RPM measurement depended heavily on the proper alignment of the IR sensor with the rotating shaft. Moreover, the system was suitable only for moderate RPM ranges and not for very high-speed applications. Another drawback was the requirement of manual scaling, since the counter value had to be multiplied by 10 to obtain the RPM. In addition, the timing of the 555 monostable circuit used for pulse filtering had to be manually adjusted for different ranges of RPM, which reduced the flexibility of the system.

V. CONCLUSION AND FUTURE WORK

This project successfully demonstrated a digital RPM meter using simple logic ICs and timers. The

system accurately measured RPM by counting pulses within a 6-second time frame. Future improvements could include automatic scaling circuits, higher-resolution sensors for wider RPM ranges.

Refreances

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